
Null-model treatment of the sensory-motor boundary changes an evolutionary connectome comparison

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Abstract

Randomised copies of a connectome are the usual baseline for asking whether measured wiring matters, and the answer depends on what the randomisation preserves. We evolved embodied foraging agents whose brains are a compressed adult *Drosophila* connectome (FlyWire v783; 512 cell-type groups and 1,000 Kenyon cells) alongside agents built on randomised wiring, in pre-registered experiments with ten seeds, four ecologies and 600 generations. Two standard randomisations, a column shuffle and degree-preserving edge swaps, route 10.6 to 10.7 % of olfactory output directly onto descending motor groups, against 0.012 % in the connectome. On the registered primary endpoint, fitness averaged over the run, no difference was detected; at the last common-garden probe the connectome was behind both controls (−0.22 and −0.20 fitness units on seed means). Against controls that keep every sensory-output and motor-input edge and rewire only the interior, the seed-mean difference lay within a ± 0.10 equivalence bound (+0.002 and −0.074, unchanged under a calibration that also matches activity spread), although per ecology the interior column shuffle was ahead by 0.26 in one of four ecologies at ten seeds, a lead that ten further pre-registered seeds did not replicate. Rewiring the connectome so that it acquires the shortcut raised its fitness by 0.44 (10 of 10 seeds) and its dependence on olfaction from 0.15 to 0.99; graded doses raised both in step; at comparable swap counts the full dose was ahead of an interior-only sham by 0.53 (10 of 10 seeds); and a sham that rewired the same boundary edges without creating shortcuts matched the connectome (+0.007) while the full dose was ahead of it by 0.60. What a null preserves at the sensory-motor boundary can decide an evolutionary connectome comparison, and sensory-to-motor path statistics belong next to the degree statistics a null is said to preserve.

1 Introduction

Whole-brain wiring diagrams of the adult fly are public [Dorkenwald et al., 2024; Schlegel et al., 2024], and models constrained by them reproduce feeding circuits [Shiu et al., 2024], visual responses [Lappalainen et al., 2024], head-direction dynamics [Duan et al., 2025] and locomotion [Vaxenburg et al., 2025]. Whether the measured wiring is better than chance is usually asked by comparing it with a randomised network that preserves some of its statistics. The verdict of such a comparison is a statement about the properties the randomisation changed [Váša & Mišić, 2022], so the choice of null decides what is being tested.

Our first pre-registered experiment found that randomised wiring out-evolves a compressed fly connectome. Following that result up showed that the two standard randomisations we had used move about a tenth of olfactory receptor output directly onto descending motor neurons, a class of connection that carries 0.012 % of calibrated olfactory output in the connectome, where 52 of 53 descending

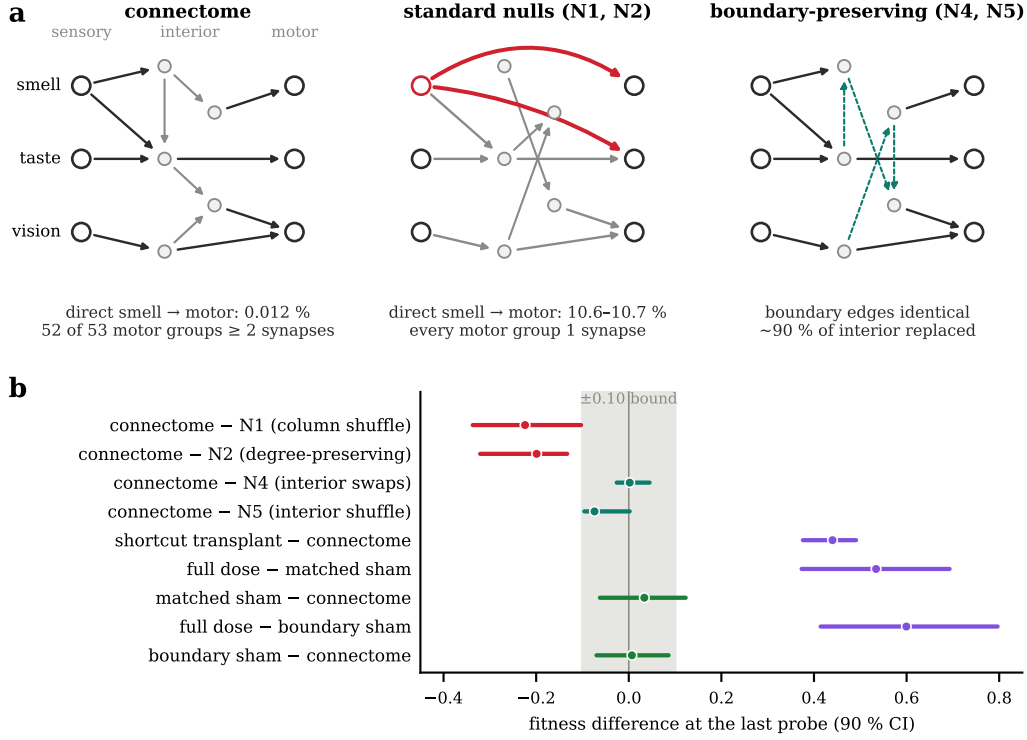


Figure 1: What the null preserves at the sensory-motor boundary, and what it changes. (a) In the compressed connectome olfactory input reaches descending (motor) groups through several synapses; the standard column-shuffle and degree-preserving nulls route about a tenth of olfactory output directly onto them; the boundary-preserving nulls keep every sensory-output and motor-input edge and replace about nine in ten interior edges. (b) Differences in evolved fitness at the last common-garden probe with 90 % intervals: connectome minus each null on seed means over four ecologies (median), and the shortcut interventions over two ecologies (transplant: median; full dose minus sham and sham minus connectome: mean, as registered). The last two rows are the boundary-targeted sham (Section 4.5), from separate runs with their own connectome island. The grey band is the ± 0.10 equivalence bound.

groups are two or more synapses from olfactory input. This is not an error in those nulls; it is what randomising the wiring between sensory and motor neurons does. But in an embodied task where a short reactive route is useful, it changes what the comparison measures. This paper reports what happens when the null keeps the sensory-motor boundary fixed, and what happens when the shortcut is moved into the connectome and out of the control (Figure 1).

Contributions:

- **A measurement** of how two standard randomisations change the sensory-to-motor boundary of a connectome-derived agent: the share of olfactory output reaching descending groups directly rises from 0.012 % to 10.6 to 10.7 %, and the median path length from olfactory receptors to descending groups falls from three synapses to one.
- **A pre-registered comparison** against boundary-preserving controls, which keep every edge out of a sensory group and every edge into a motor group: the connectome’s end-of-run disadvantage shrinks to within a ± 0.10 equivalence bound on seed means, with the per-ecology heterogeneity and the null primary endpoint reported alongside.
- **Pre-registered interventions** that move the shortcut: shortcut-creating rewiring raises the connectome’s evolved fitness and transfers the controls’ olfactory strategy to it; shortcut-removing rewiring costs a control its advantage where it had one; the effect rises with dose; an interior-only sham with a comparable number of swaps does not reproduce it; and neither does a sham that rewires the same boundary edges without creating shortcuts.

- **Open protocols, runs and a dated research log**, including a retraction and a correction made after pre-registration, so that each result can be traced to the protocol that preceded it.

2 Related work

Connectome-constrained models. Fly connectome models are usually evaluated on how well they reproduce measured activity or behaviour [Shiu et al., 2024; Lappalainen et al., 2024; Duan et al., 2025]. We instead ask whether measured wiring is a better starting point for evolution than randomised wiring, a question closer to Worrell et al. [2017], who found faster sensor-to-effector spreading in the fly connectome than in degree-preserving nulls.

Null models for connectomes. A null keeps some properties and randomises the rest, and constraining it to the property under test is standard advice [Váša & Mišić, 2022]; degree-preserving rewiring misses spatially constrained structure [Salova & Kovács, 2025]. Dhiman [2026] found that a fly visual-system model’s advantage over degree-preserving nulls largely disappears under a shared random initialisation, a verdict change through initialisation and degree effects rather than path structure. Cho [2026] released a behavioural benchmark for whole-connectome simulations with shuffled and degree-preserving controls. We apply the constrain-the-null advice to the sensory-motor boundary and show a case where it changes the outcome.

Architecture, randomisation and measurement in machine learning. Architecture alone can encode useful behaviour [Gaier & Ha, 2019], and which structure a randomised baseline keeps determines what a comparison can attribute to it [Zhou et al., 2019]. Randomisation tests are only informative about the properties they destroy [Adebayo et al., 2018], and an apparent effect can come from the choice of measurement rather than from the system [Schaeffer et al., 2023]. With ten seeds per condition we follow the recommendation to report interval estimates over runs rather than point estimates [Agarwal et al., 2021].

3 Methods

All quantities are dimensionless simulation units. Full specifications are in Appendix A (brain, dynamics, world, evolution) and Appendix B (controls).

3.1 Agents

From FlyWire v783 we drop optic-lobe, visual-centrifugal and photoreceptor classes, group the remaining neurons by cell type $\times$ hemisphere and keep $K = 512$ groups, starting with a mandatory set of olfactory, gustatory, mushroom-body, dopaminergic, looming and named descending groups and filling the rest by a sensory-to-descending path score. Signed weights come from synapse counts and predicted neurotransmitters [Eckstein et al., 2024]; the matrix has 14,989 non-zero entries, and sensory rows are zeroed so that sensors act as transducers. Group rates follow leaky rectified-tanh dynamics. 1,000 Kenyon cells keep their measured projection-neuron inputs, with APL-like inhibition holding 7 to 9 % active, and a dopamine-gated plastic Kenyon-cell-to-output matrix resets at birth. Forward speed, turning and eating are read out from 24 bilateral descending-neuron types, initialised from named descending neurons (e.g. DNa01/DNa02 for turning), with a per-brain innate feeding reflex. The compression keeps named sensory and motor pathways on purpose, so the conditions compared share this boundary and differ in the wiring between its two sides.

3.2 World, ecologies and evolution

A unit square holds 24 odor-marked patches (nutritious, potentially toxic, neutral), with the odor-to-role assignment redrawn every lifetime of 400 steps and an energy budget that forces foraging. Four ecologies cross a predator that is stationary (P0) or pursuing (P1) with potentially toxic patches that are all nutritious (T0) or all toxic (T1). Each wiring condition evolves as its own island of 512 agents for 600 generations (tournament selection, 4 % elitism); mutations perturb weights, group sizes, leaks, biases, gains, learning rates and read-outs, and can add or prune edges.

3.3 Controls and interventions

Controls are drawn per seed and shared across that seed’s ecologies (Table 1). N1 moves each presynaptic column’s non-zero weights to random non-sensory rows; N2 performs Maslov–Sneppen target swaps [Maslov & Sneppen, 2002] with $10 \cdot |E|$ attempts; N4 and N5 apply the same operations to *interior* edges only, whose source is not sensory and whose target is not a descending group. The engine aborts unless N4 and N5 boundary edges are bit-identical to the connectome; they replace about nine in ten interior edges (interior overlap 0.111 to 0.121 for N4 and 0.089 to 0.095 for N5 over the 40 constructions of the corrected grid). The Kenyon-cell input matrix is randomised by degree-preserving swaps in N2, N4 and N5 and by a column shuffle in N1.

Table 1: Randomised controls. All keep edge counts and Dale signs.

code	operation	additionally preserved
N1	column shuffle	out-degree, per-source weight multiset
N2	degree-preserving target swaps	in- and out-degree
N4	N2 on interior edges only	as N2, plus every sensory-output and motor-input edge
N5	N1 on interior edges only	as N1, plus every sensory-output and motor-input edge

Three interventions move the shortcut directly. **AS** adds it to the connectome: pairs of edges, one from an olfactory receptor group to a non-motor target and one from a non-sensory source to a descending group, exchange targets, which keeps every degree and each weight with its source, until the olfactory shortcut share (the fraction of olfactory output weight landing directly on descending groups) reaches that of N2 (0.106). **N2R** runs the same operation in reverse on N2 until its share is 0. A **dose** series raises the connectome’s share to at least 1, 3, 5 and 10 %, and a **sham** performs 101 degree-preserving swaps on edge pairs whose source is not an olfactory receptor and whose target is not a descending group, leaving the shortcut share at 0.0001. The sham does not touch the boundary edges that the dose rewires; §4.4 discusses what it therefore does and does not control.

3.4 Calibration

Every network is row-normalised and scaled by a global and a Kenyon-cell gain found by bisection so that mean group activity is 0.15 and mean Kenyon-cell activity 0.05 over a fixed stimulus set. This matches mean activity but not its spread: the connectome’s median between-group activity SD is 0.163 with 46 silent groups of 512, against medians of 0.080 to 0.087 and 8 to 19 silent groups for the controls (N1 to N3 in the first grid, N4 and N5 in the corrected grid). Row normalisation also means that a boundary edge bit-identical before calibration can carry a slightly different share of weight after it; all path statistics below are computed on the calibrated matrices the runs used. A pre-registered re-run of the corrected grid under a calibration that also matches the spread of activity (between-group SD 0.118 and 3 to 7 silent groups in every condition) returned the same verdicts (Section 4.2, Appendix F).

3.5 Evaluation and statistics

Every 50 generations the 16 agents with the highest training fitness per island are evaluated for 8 further lifetimes in a common garden: the random-number generator is reset before the batch so that all conditions and ablations face the same generated worlds. The registered **primary endpoint** is this fitness averaged over generations 0 to 550 (trapezoid rule); the registered **secondary endpoint** is the fitness at the last probe, generation 550. Olfaction, vision and plasticity are ablated in parallel copies of the batch.

Seeds are the unit of replication. Paired differences (connectome minus control) are tested with two-sided Wilcoxon signed-rank tests, Holm-corrected across controls, and summarised by medians with percentile bootstrap intervals over seeds (10,000 resamples). Runs of one seed share their controls across ecologies, so cross-ecology statements average each seed’s ecologies first and test the ten seed means. For absence of a difference we require equivalence: the 90 % interval of the median paired difference must lie inside $\pm\delta$ with $\delta = 0.10$. The bound was set after the first grid, when the shortcut

effect (at least 0.20) was known, and before any equivalence interval was computed; Appendix F reports how each verdict depends on δ .

3.6 Pre-registration and a correction

Each experiment below was run under a protocol frozen before its data (listed in Appendix D), with engine hashes recorded. The first grid used a heading-noise term with a non-zero mean that no network could offset; it was found through a question raised in AI-assisted review of a draft and corrected, and every later experiment uses zero-mean noise. The first grid is reported because it motivated the rest; its generation-0 result is not used (Appendix D).

4 Results

4.1 Standard controls carry sensory-to-motor shortcuts and out-evolve the connectome

In the connectome, exactly two of 14,989 edges run from an olfactory receptor group to a descending group, carrying 0.012 % of calibrated olfactory output; one descending group is one synapse from olfactory input, 15 are two, 36 are three and one is four. A column shuffle or degree-preserving swaps put 10.6 to 10.7 % of olfactory output directly onto descending groups and make every descending group reachable in one synapse (Table 2). The boundary-preserving controls keep the connectome’s structure; their small differences in the direct shares come from calibration. For vision, whose routes are already short, randomisation works the other way and lowers the direct share.

Table 2: Sensory-to-motor structure of the calibrated networks (medians over 10 seeds). “Direct”: share of that sense’s output weight landing on descending groups; “hops”: median shortest path from that sense to a descending group.

condition	smell: direct	smell: hops	vision: direct	taste: direct
connectome	0.00012	3	0.381	0.121
N1 column shuffle	0.107	1	0.108	0.113
N2 degree-preserving	0.106	1	0.128	0.093
N4 interior swaps	0.00010	3	0.340	0.125
N5 interior shuffle	0.00011	3	0.369	0.123

In the first pre-registered grid (4 ecologies $\times$ 10 seeds, with the uncorrected noise term), connectome populations were overtaken: at the last probe the column shuffle was ahead in all ten seeds on seed means ($p_{\text{holm}} = 0.0059$) and the degree-preserving control in eight ($p_{\text{holm}} = 0.0195$), while connectome populations depended far less on olfaction (Appendix E).

4.2 Against boundary-preserving controls the seed-mean difference lies within the equivalence bound

We re-ran the grid with the noise corrected and with N4 and N5 added (40 runs, 5 conditions each). On the registered primary endpoint no difference was detected against any control (seed-mean medians -0.03 for N1, -0.02 for N2, $+0.03$ for N4 and -0.01 for N5, none significant); the effects below emerge late in evolution and are measured on the registered secondary endpoint. At the last probe the connectome was behind the shortcut-carrying controls ($p_{\text{holm}} = 0.029$ for N1 and 0.016 for N2 on seed means), and against the boundary-preserving controls the seed-mean difference lay inside the ± 0.10 bound: $+0.002$ for N4 and -0.074 for N5 (Table 3). The N4 verdict holds for any bound down to 0.05; the N5 verdict holds at 0.10 but not at 0.075 and is marginal.

Table 3: Corrected grid, last probe (generation 550), connectome minus control: median over 10 seeds, with the 90 % interval below. Equivalence is assessed only on seed means across ecologies.

control	seed means	P0T0	P1T0	P0T1	P1T1
N1 (shortcut)	-0.224	-0.621	-0.312	-0.159	$+0.054$

control	seed means	P0T0	P1T0	P0T1	P1T1
N2 (shortcut)	[-0.34, -0.10] -0.199	[-0.66, -0.47] -0.403	[-0.44, -0.22] -0.338	[-0.21, +0.01] -0.017	[-0.05, +0.13] +0.015
N4 (boundary)	[-0.32, -0.13] +0.002	[-0.42, -0.31] -0.064	[-0.53, -0.07] +0.021	[-0.33, +0.01] -0.031	[-0.13, +0.14] +0.120
N5 (boundary)	[-0.026, +0.045] -0.074 [-0.096, +0.002]	[-0.26, +0.00] -0.264 [-0.37, -0.18]	[-0.03, +0.04] +0.067 [-0.07, +0.09]	[-0.12, +0.09] -0.015 [-0.10, +0.06]	[-0.06, +0.28] +0.070 [-0.01, +0.21]

The seed-mean equivalence does not hold ecology by ecology, and we do not claim it there. In the safe-food ecology (P0T0) the interior column shuffle N5 is ahead of the connectome by 0.26 ($p_{\text{holm}} = 0.029$), more than twice the bound; in the predator-plus-toxin ecology the connectome is ahead of N4 by 0.12 with an interval spanning zero. We then ran seeds 10 to 19 in the two T0 ecologies under rules fixed before those runs (Appendix F). N5's lead in P0T0 did not replicate: in the new seeds A – N5 was +0.031 (90 % CI -0.068 to +0.152), and over all 20 seeds -0.104 (-0.208 to +0.016, $p_{\text{holm}} = 0.27$). At $n = 20$ the per-ecology intervals for N4 and N5 lie inside ± 0.10 in P1T0 (+0.003 and +0.056) but not in P0T0 (N4 -0.011, -0.114 to +0.080). N1 stays ahead in both ecologies (-0.491 and -0.333, $p_{\text{holm}} \leq 0.004$); N2 stays ahead in P1T0 (-0.203, $p_{\text{holm}} = 0.046$), but its P0T0 lead shrank to -0.214 (-0.375 to +0.042, $p_{\text{holm}} = 0.25$), and in seeds 10 to 19 alone the connectome was ahead of N2 there (+0.100). The larger sample therefore weakens both the one per-ecology lead of a boundary-preserving null and part of the standard nulls' per-ecology advantage; the seed-mean contrast over four ecologies, which the extension did not cover, is unchanged.

The sensory strategy follows the same split. Blocking olfaction costs the connectome 0.03 against 0.68 for N1 and 0.64 for N2 ($p_{\text{holm}} = 0.008$), and 0.10 for N4 and 0.17 for N5; the connectome's differences from N4 and N5 (-0.07 and -0.11) are five- to ninefold smaller than those from N1 and N2 and not significant after correction ($p_{\text{holm}} = 0.13$ for both).

Because global calibration leaves the connectome with a wider spread of activity than the controls, we re-ran the corrected grid (40 runs) under a distribution-matched calibration that equalises mean, spread and silent fraction across conditions, with verdict rules fixed beforehand. Every run passed the manipulation check (control SD within 0.01 and silent-group count within 10 of the connectome's). The verdicts held: on seed means the boundary-preserving controls stayed inside ± 0.10 (A – N4 +0.055, 90 % CI +0.035 to +0.097; A – N5 -0.009, -0.076 to +0.046) and the standard controls stayed ahead (A – N1 -0.274, -0.306 to -0.123; A – N2 -0.146, -0.231 to -0.037). The calibration shifted the N2 contrast by +0.101 (+0.016 to +0.133) and left the other three shifts undetermined, and N5's lead in P0T0 did not appear under it either (-0.066, -0.320 to +0.124; Appendix F).

4.3 Moving the shortcut in and out

The pre-registered intervention (20 runs in the two ecologies with the clearest overtaking) rewires the shortcut in both directions while keeping degrees and Dale signs (Figure 2). Adding shortcuts to the connectome raised its fitness at the last probe in every seed, by +0.44 on seed means (10 of 10 seeds, $p = 0.002$; +0.445 in P0T0 and +0.475 in P1T0), and its time-averaged fitness by +0.26 (10 of 10, $p = 0.002$). The effect grew over evolution: +0.04 at generation 0 ($p = 0.32$; the interval is wider than the equivalence bound, so an initial effect is not excluded) and +0.40 from generation 0 to the last probe (10 of 10 seeds, $p = 0.002$). The intervention also transferred the sensory strategy, raising the cost of blocking olfaction from 0.15 to 0.99 ($p = 0.002$), close to the degree-preserving control's 0.84.

Removing the shortcut from the degree-preserving control cost it 0.43 fitness in the safe-food ecology ($p = 0.004$) and an estimated 0.00 in the predator ecology (difference between ecologies -0.36, $p = 0.027$); in neither ecology could the shortcut-free control be distinguished from the connectome, and its olfactory dependence fell from 0.84 to 0.58 ($p = 0.027$). Both interventions also rewire edges that are not shortcuts, so this experiment shows that *shortcut-creating rewiring* is sufficient for the advantage, not that the shortcut edges alone are.

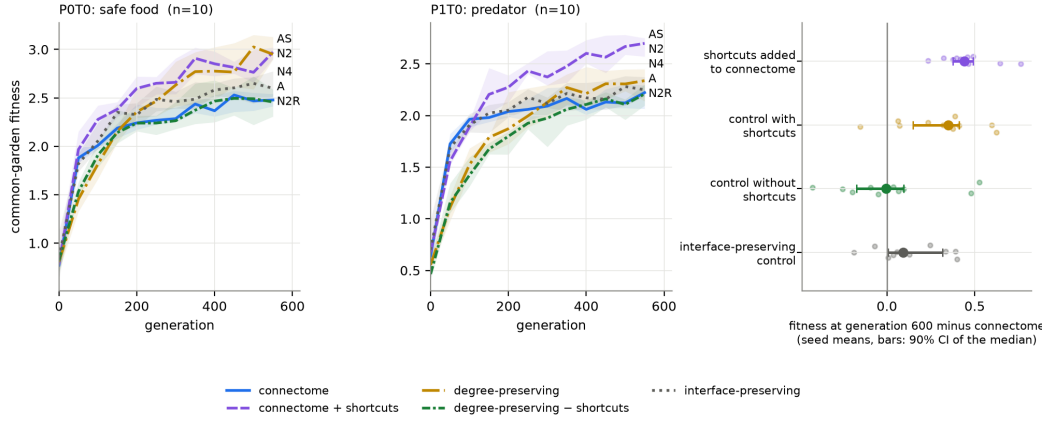


Figure 2: Intervention. Left and middle: fitness trajectories in the two ecologies (median and interquartile range over 10 seeds). Right: differences from the connectome at the last probe, seed means over both ecologies, with 90 % intervals. Adding shortcut-creating rewiring (AS) reproduces the control's advantage; removing it from the degree-preserving control (N2R) removes the advantage where it was clear.

4.4 Graded doses and an interior-only sham

To ask whether the effect follows the amount of shortcut, a pre-registered dose series raised the connectome's olfactory-to-descending share to at least 1, 3, 5 and 10 % (2 ecologies $\times$ 10 seeds). Fitness rose across doses, on seed means 2.37, 2.54, 2.69, 2.75 and 2.93 from the connectome to the 10 % dose (Page's L, $z = 4.68$, $p = 1.4 \times 10^{-6}$; per-seed Spearman correlation positive in 9 of 10 seeds), in both ecologies ($p = 1.5 \times 10^{-5}$ and 0.004), and dependence on olfaction rose with it ($p = 3.4 \times 10^{-5}$ and 5.6×10^{-7} ; Figure 3). At generation 0 no trend was detected across doses ($p = 0.98$).

The full dose took 68 to 129 swaps depending on the seed (mean 98.5), comparable to the sham's 101, and was ahead of it by 0.534 on seed means (90 % CI +0.373 to +0.693, 10 of 10 seeds; +0.525 in the safe-food and +0.543 in the predator ecology). It was ahead in all seven seeds in which it used fewer swaps than the sham. The number of degree-preserving swaps therefore does not account for the effect. What the sham does not control is *where* the swaps land: the dose necessarily rewires olfactory-output and descending-input edges, while the sham avoids them, so this comparison separates shortcut-creating rewiring from interior-only rewiring, not the shortcut edges from other changes at the boundary. Section 4.5 closes that gap with a sham that rewires the same boundary edges. In this dose series shams were run only at the full swap count; the boundary-targeted experiment pairs a sham with every dose.

The sham itself was registered to fall inside the equivalence bound against the connectome and did not: +0.034 on seed means (90 % CI -0.062 to +0.123), ahead by 0.144 in the safe-food ecology (interval +0.072 to +0.222) and behind by 0.077 in the predator ecology (interval spanning zero). Interior-only rewiring is therefore not demonstrably inert, but its effect is small next to the dose's and does not keep its sign across ecologies (Appendix G).

4.5 A boundary-targeted sham separates the shortcut from boundary rewiring

A shortcut swap turns $ORN \rightarrow X$ and $Y \rightarrow DN$ into $ORN \rightarrow DN$ and $Y \rightarrow X$. The boundary-targeted sham BSk rewires the same two boundary edges without creating a shortcut: it adds an interior edge $Z \rightarrow W$ and rotates the targets, giving $ORN \rightarrow W$, $Y \rightarrow X$ and $Z \rightarrow DN$. The olfactory weight lands in the interior instead of on a descending group, and the descending group receives Z's weight instead. Each seed performs as many rotations as its own ASk performed swaps. $Z \rightarrow W$ is the one of 64 random interior edges whose weight is closest in magnitude to $ORN \rightarrow X$ (median ratio 0.92 to 1.05 across constructions). Degrees, signs and the olfactory-to-descending share (0.0001) are unchanged, which the engine checks. Nine islands (A; AS1, AS3, AS5, AS10; and their shams BS1, BS3, BS5, BS10) evolved together in the two ecologies over ten seeds, under a protocol frozen before the runs

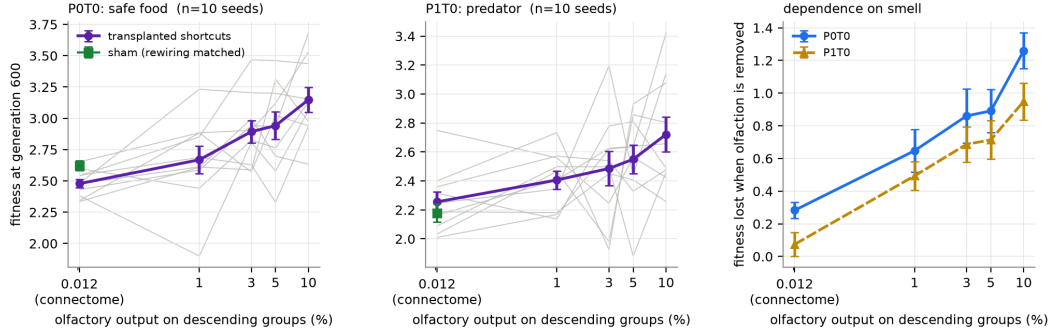


Figure 3: Graded shortcut transplant. Left and middle: fitness at the last probe against the share of olfactory output landing directly on descending groups, per ecology; grey lines are seeds, purple the mean with standard error, and the green square the interior-only sham (101 swaps) plotted at the connectome’s own share. Right: fitness lost when olfaction is blocked, against the same axis.

(PROTOCOL_BSHAM.md). The primary contrast is AS10 – BS10 at the last probe, with a 90 % bootstrap interval of the seed mean: *shortcut-specific* if the lower bound exceeds +0.10, *boundary rewiring suffices* if the interval lies inside ± 0.10 .

The verdict was shortcut-specific. The full dose was ahead of its boundary-targeted sham by 0.600 (90 % CI +0.414 to +0.797), and by 0.568 and 0.631 in the two ecologies. The registered robustness check re-evaluated the generation-599 elites in four newly generated worlds and gave the same category (+0.639, +0.490 to +0.792). The sham itself was indistinguishable from the connectome (BS10 – A = +0.007, -0.070 to +0.086, inside ± 0.10). The gap between each dose and its matched sham grew with dose (0.16, 0.28, 0.36 and 0.60 for 1, 3, 5 and 10 %; Page’s L, $z = 3.18$, $p = 7.4 \times 10^{-4}$). Blocking olfaction cost the full dose 1.18 and its sham 0.35. The transplant’s gain over the connectome replicated (AS10 – A = +0.606, +0.435 to +0.798). Rewiring the boundary edges the same number of times, without creating direct olfactory-to-descending connections, therefore reproduced none of the effect: it follows the shortcut edges themselves. Three runs failed with GPU out-of-memory (one of them twice), one completed attempt was deleted by a driver error and one attempt was stopped to test a memory fix; every affected run was re-run from scratch with identical flags before any outcome was inspected. Sixteen of the 20 runs used an engine copy that differs only in forcing evaluation earlier to save memory and was bit-identical to the original on the CPU (Appendix G).

4.6 Further checks

Zeroing interior weights of evolved populations cost 18 to 23 % of fitness and permuting them 46 to 55 %, in every condition, with no detected difference between the connectome and any control (all $p \geq 0.16$). The evolved agents are therefore sensitive to their interior wiring rather than running on the boundary alone; this is perturbation sensitivity, not evidence that the task requires interior computation. In the first grid, the fraction of one-step mutants beating their parent’s clones was similar for connectome and control genomes (9.6 % against 9.4 and 9.8 % at generation 0), and connectome genomes lost more to large mutations (Appendix E). We detected no odor-specific associative learning in any condition, and over evolution weights moved by 0.5 to 3 % of their initial spread while gains, leaks and read-outs moved by about 6 to 11 initial standard deviations (results/evolution_analysis.txt, the three calibration schemes of paper-v1) (Appendix H).

5 Discussion

In this model the connectome’s standing depended on how the randomised control treated connections between sensors and motor neurons. The two standard nulls randomised the sensory-motor boundary along with everything else, so the comparison they support is about the boundary as much

as about the interior wiring we meant to test. With the boundary held fixed, the end-of-run seed-mean difference fell within ± 0.10 ; moving the shortcut into the connectome reproduced the controls' advantage and olfactory strategy; and the advantage followed the dose. A sham that rewired the same boundary edges the same number of times without creating shortcuts left fitness where the connectome was, so the effect is attributable to the direct olfactory-to-descending connections rather than to rewiring at the boundary as such.

Which properties a null should preserve depends on the hypothesis. A randomisation meant to test whether interior wiring matters should not change which sensors reach which effectors in one step; one meant to test the whole architecture may. Keeping every sensory-output and motor-input edge, as N4 and N5 do, is one way to hold the boundary fixed, and reporting calibrated sensory-to-motor path statistics next to degree statistics makes the choice visible. The same concern applies wherever a task rewards short reactive routes; whether it matters in static benchmarks depends on their tasks, and the statistics are cheap to report either way. Together with Dhiman [2026], who found a verdict change through initialisation and degree effects, this suggests that which null was used, and what it preserved, matters more for connectome comparisons than has been usual.

What we can say about the connectome itself is narrow. On seed means over four ecologies we found no end-of-run advantage for measured interior wiring over boundary-preserving controls, within ± 0.10 ; that is a bounded null for one estimand, not proof of no effect, and the one per-ecology lead of the interior column shuffle did not replicate in ten further seeds. Per-ecology estimates are also less stable than the seed means: the degree-preserving control's lead in the safe-food ecology, which motivated the intervention there, shrank to an interval spanning zero at twenty seeds. Our agents are a compressed rate model in a two-dimensional world, no condition evolved detectable associative learning, and evolution adapted gains and read-outs far more than weights. A task that requires what fly circuits are built for, such as learning from experience within a lifetime, could give a different answer, and is the natural next test.

Limitations. One connectome, one compression and one family of two-dimensional worlds; one mutation operator, which tunes far more than it rewires; a toxin factor confounded with food abundance and a stationary rather than absent predator; global calibration in the main experiments (verdicts unchanged under distribution matching); a common-garden evaluation that reuses one generated world stream per probe; population samples rather than elites in the post hoc assays; equivalence bounded at ± 0.10 on seed means with ten seeds and only where stated (twenty in two ecologies), with the bound set after the first grid; interventions that rewire more than the shortcut edges (the boundary-targeted sham controls this at the boundary, not everywhere), an interior-only sham that failed its own registered prediction; and boundary edges that mutation can add during evolution.

Data and code availability

Engines, analysis and figure scripts, frozen protocols, the dated research log and run outputs are at <https://github.com/gyujeongion/flyconnectome-nulls>; an archived copy with the manuscripts is at <https://doi.org/10.5281/zenodo.22871090>. Raw per-generation logs and stored genomes (about 2 GB) are not included; the analysis outputs computed from them are. Re-running a seed does not repeat a run bit for bit: after every common-garden probe, which runs under a fixed random seed, the engines reseed the evolutionary random stream from the clock, and the GPU scatter used in mutation does not fix which write survives when two mutations hit the same weight. A re-run reproduces the null constructions, calibration and generation 0 exactly and later generations only in distribution; seeds are therefore independent replicates, not reproducible trajectories. FlyWire v783 connectivity and annotations are public. Every run used one Apple M1 Ultra with MLX (about 36 minutes per 5-condition, 600-generation run).

Use of AI tools

The engines, analysis and figure scripts were written, and this manuscript drafted, with an AI coding assistant (Claude Code, Anthropic); two further language models critiqued drafts. Every number is regenerated from stored run outputs by scripts in the repository, and the author directed the study and is responsible for the text (Appendix J).

Funding and conflicts of interest

This research received no external funding. The author declares no conflicts of interest.

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A Model details

A.1 Brain model

Compression. From FlyWire v783 we drop optic-lobe, visual-centrifugal and photoreceptor classes, group the remaining neurons by cell type $\times$ hemisphere (mushroom-body output, dopaminergic and MBIN types merged across hemispheres), and keep $K = 512$ groups. A mandatory set is kept first: 16 olfactory receptor-neuron types including the CO_2 channel ORN_V; sugar and bitter gustatory receptor neurons; every antennal-lobe projection-neuron group receiving at least 5 % of its input from kept ORNs plus the 24 most ORN-driven local-neuron groups; mushroom-body output, dopaminergic and modulatory neurons; APL and DPM; the looming detectors LC4 and LPLC2; the descending neurons DNa01, DNa02, DNp09, MDN, DNp01, DNgl1, DNgl2_a; the 48 highest-scoring groups on a gustatory-to-dopaminergic path score; PFL3 output targets and the path from them to DNa01/DNa02. Remaining slots go to the highest scores on

$$\text{score}(i) = f(i) \cdot b(i), f = \sum_{h=1..4} (T^h \mathbf{1}_{\text{sensory}})(i), b = \sum_{h=1..4} ((T^T)^h \mathbf{1}_{\text{descending}})(i), T = D^{-1} S_{\text{abs}},$$

with S_{abs} the group-level absolute synapse-count matrix and D its row sums. Weights are $W^0_{ij} = S_{\text{signed},ij} / (\sum_k S_{\text{abs},ik} + 1)$ with neurotransmitter-predicted signs (Eckstein et al., 2024). Sensory rows are zeroed, so sensors act as transducers. The resulting matrix has 14,989 non-zero entries.

The compression keeps named sensory and motor pathways on purpose, and the read-outs start from named descending neurons, so the comparison is between agent designs that share this connectome-derived boundary and differ in the wiring between the two sides of it.

Dynamics. With group rates $r \in \mathbb{R}^K$, size factors $c = \exp(s)$, leak $\alpha = \sigma(a - 1)$, bias b , sensor vector $S \in \mathbb{R}^{40}$ with gains g_{in} and fixed projection B_{in} from sensor channels onto their receptor groups, Kenyon-cell rates k , plastic $\text{KC} \rightarrow \text{MBON}$ matrix W_{km} , fixed $\text{KC} \rightarrow \text{group}$ matrix W_{ko} and MBON one-hot map P_{MB} :

$$x = W(r \odot c) + W_{\text{ko}} k + P_{\text{MB}}(W_{\text{km}} k) + B_{\text{in}}(S \odot g_{\text{in}}) + b \leftarrow (1 - \alpha) \odot r + \alpha \odot \tanh(\max(x, 0))$$

1,000 Kenyon cells keep their measured projection-neuron inputs W_{pk} (10.3 non-zero group inputs each). With $u = g_{\text{kc}} \cdot W_{\text{pk}}(r \odot c)$, $k = \tanh(\max(u - (\text{mean}(u) + \lambda \cdot \text{SD}(u)), 0))$ with heritable λ , an APL-like inhibition keeping 7 to 9 % of Kenyon cells active. Plasticity is dopamine-gated: $d = M_{\text{DAN}} r_{\text{DAN}}$, $\bar{d} \leftarrow 0.95 \bar{d} + 0.05 d$, $W_{\text{km}} \leftarrow \text{clip}(W_{\text{km}} + \eta \odot (d - \bar{d}) k^T, 0, 2)$, with heritable per-MBON rates η and measured $\text{DAN} \rightarrow \text{MBON}$ connectivity. Plastic weights reset at birth.

Read-outs and actions. For the 24 descending-neuron types present on both sides, with rates D_{L} and D_{R} : forward = $R_0 \cdot (D_{\text{L}} + D_{\text{R}}) + R_{\text{M},0} r_{\text{MBON}} + b_0$; turn = $R_1 \cdot (D_{\text{L}} - D_{\text{R}}) - m$ with $m \leftarrow 0.9 m + 0.1 (D_{\text{L}} - D_{\text{R}})$; eat = $R_2 \cdot (D_{\text{L}} + D_{\text{R}}) + R_{\text{M},1} r_{\text{MBON}} + b_2$. R_0 and R_1 start from named descending neurons (forward 2·DNp09 – 2·MDN + DNp01; turn 2·DNa01 + 2·DNa02) scaled by 8 plus $\mathcal{N}(0, 0.3)$; R_2 is an innate feeding reflex derived per brain from its own descending responses to sugar, bitter and odor. Speed is $v = v_{\text{max}} \sigma(\text{forward})$, reduced to 0.2 while eating; the agent eats when $\text{eat} > 0$; $v_{\text{max}} = 0.012$, $\omega_{\text{max}} = 0.35$. Heading follows

$$\theta \leftarrow \theta + \omega_{\text{max}} \tanh(\text{turn}) + 0.05 (\xi - 0.5), \xi \sim U(0, 1).$$

The first grid (Appendix E) used 0.05 ξ instead, which adds a constant 0.025 rad per step; Appendix D gives the consequences. All later experiments use the zero-mean form above.

A.2 World

A unit square with reflecting walls holds 24 patches: 10 nutritious, 8 potentially toxic, 6 neutral, each carrying one of 8 odors as a 15-dimensional glomerular vector, with the odor-to-role assignment redrawn every lifetime. Antennae sit 0.05 from the body at ± 0.6 rad; odor concentration is $\sum_p \exp(-d^2/2\sigma^2)$ with $\sigma = 0.06$; predator odor enters the CO_2 channel with $\sigma_p = 0.12$; looming input to LC4/LPLC2 is $\text{clip}(0.04/d, 0, 1)$ on the predator’s side when it is in the frontal field. Eating radius 0.055; a nutritious bite gives +0.04 energy, a toxic bite costs $g_{\text{bad}} = 0.25$ and triggers a delayed malaise signal reaching the bitter channel, and toxic food tastes sweet on contact. Nutritious patches hold 12 bites, toxic ones 1; depleted patches respawn elsewhere. Energy starts at 0.7 and falls by $0.0025 + 0.0008 \cdot v/v_{\text{max}} + C_{\text{brain}}$ per step, with $C_{\text{brain}} = 0.1 \cdot (0.0006 \cdot N_{\text{eff}}/N_{\text{ref}} +$

$0.0004 \cdot n_edges / n_edges,ref$), where N_eff is the neuron count implied by the evolved group sizes, $N_ref = 13,718$ the count of the initial brain, which is the 8,541 neurons of the 512 groups plus a fixed offset of 5,177 for the Kenyon-cell population, which does not evolve, n_edges the number of non-zero weights and $n_edges,ref = 14,989$; an agent that never eats dies before its lifetime ends. Fitness is

$$fitness = 0.04 \cdot (\text{nutritious bites}) - g_bad \cdot (\text{toxic bites}) - C_brain \cdot (\text{steps alive}) + 0.3 \cdot (\text{alive at end}) + 0.0005 \cdot (\text{steps alive}),$$

so the brain cost is charged twice by design, once through energy and once directly.

Ecologies. Predator: stationary and harmless (P0), or pursuing at $0.2 \cdot v_max$ within 0.45 and killing within 0.035 (P1). In P0 the predator stays put and still emits CO_2 and looming cues; only capture is disabled. Toxin: the 8 potentially toxic patches are all nutritious (T0, 18 nutritious patches) or all toxic (T1, 10 nutritious patches), so the toxin factor also changes food abundance. The four cells are P0T0, P1T0, P0T1, P1T1.

A.3 Evolution

One island per wiring condition, 512 agents each, tournament selection ($k = 4$), 4 % elitism, 600 generations, no migration. Each child gets 3,000 random entries of W multiplied by $\exp(\mathcal{N}(0, 0.25))$; 3 zero entries turned into edges with Dale-correct sign and magnitude $|\mathcal{N}(0, 0.05)|$, which may fall anywhere including the boundary; 60 random entries pruned with probability 0.5; entries below 10^{-3} zeroed. Gaussian perturbations (probability, SD, clip): group size (0.05, 0.15, $[-2, 2]$), leak (0.05, 0.3, $[-4, 4]$), bias (0.05, 0.1, $[-2, 2]$), learning rates (0.2, 0.03, $[-1, 1]$), input gains (0.3, 0.2, $[0, 8]$), Kenyon-cell gain (0.3, 0.1, $[-2, 2]$), Kenyon-cell sparsity (0.3, 0.1, $[0.5, 3]$), descending read-out (0.2, 0.5, $[-12, 12]$), MBON read-out (0.2, 0.5, $[-10, 10]$), read-out biases (0.3, 0.2, $[-5, 5]$).

Every 250 generations the run stores 8 genomes per island. An indexing error makes these members of the population just after selection rather than the elites; we call them population samples and use them only in the post hoc assays of Appendix H. Their fitness relative to the top 16 of the same generation is 0.98 (connectome), 0.93 (N1) and 0.98 (N2), with paired tests against the connectome giving $p = 0.14$ and 0.65; this does not prove the sampling is unbiased, and the assays that rest on these genomes carry that caveat.

B Controls, interventions and calibration

B.1 Definitions

Controls are drawn fresh per seed from one random stream, in the order the conditions are listed, and are used in all ecologies of that seed. W_pk is randomised by the same rule; W_ko and initial $KC \rightarrow MBON$ weights are shared. The first three are in common use. The last two apply the general practice of constraining a null to the property under test (Váša & Mišić, 2022) to the sensory-motor boundary; we did not find this particular constraint in earlier connectome comparisons, but it is an application of that practice, not a new principle.

code	operation	preserved
N1	permute the row positions of each presynaptic column's non-zeros	out-degree, Dale sign, per-source weight multiset
N2	Maslov–Sneppen target swaps (Maslov & Sneppen, 2002), $10 \cdot E $ attempts	in- and out-degree, weight–source pairing, sign
N3	permute entries within each (super-class $\times$ hemisphere) block pair	connections between block pairs, block-level laterality
N4	Maslov–Sneppen swaps restricted to interior edges	as N2, plus every edge leaving a sensory group and every edge entering a descending group

code	operation	preserved
N5	column shuffle restricted to interior edges	as N1, plus the same boundary edges

Interior edges are those whose source is not a sensory group and whose target is not a descending group. For N4 and N5 the engine checks at construction that the boundary edges are bit-identical to the connectome (maximum absolute difference 0) and aborts otherwise; edge counts match the connectome exactly, and interior edge overlap with the connectome is 0.09 to 0.12 (N4 0.111 to 0.121, N5 0.089 to 0.095 over the 40 constructions of the corrected grid), so both replace roughly nine in ten interior edges. Boundary preservation constrains the first and last synapse of any sensory-to-motor route; it does not fix the lengths or identities of the interior routes between them, which is what these controls are meant to randomise.

Two further conditions manipulate the shortcut directly. **AS** is the connectome with shortcuts added: pairs of edges, one from an olfactory receptor group to a non-motor target and one from a non-sensory source to a descending group, exchange targets, which preserves every in- and out-degree and keeps each weight with its source. Swaps continue until the olfactory shortcut share reaches 0.106, the value of the degree-preserving control. **N2R** is the degree-preserving control with the same operation run in reverse until its shortcut share is 0. Both interventions rewire more than the shortcut edges themselves, so they test shortcut-creating and shortcut-removing rewiring rather than an isolated variable; The two activity statistics we monitor are unchanged in AS (between-group SD 0.160 against 0.163, 44 silent groups against 48); other aspects of the dynamics may differ.

The **olfactory shortcut share** is the fraction of the absolute output weight of olfactory receptor groups that lands directly on descending groups.

B.2 Implementation

The randomisations are implemented in `evo_fix.py` (`shuffle_control`, `null_degswap`, `null_interface_degswap`, `null_interface_colshuffle`); the dose and sham operations in `evo_dose.py`. Precisely:

- **Sensory rows.** Sensory groups receive no input in the connectome (their rows are zero). N1 places each column’s weights only in non-sensory rows; degree-preserving swaps exchange the targets of two existing edges, whose targets are never sensory, so no control gives a sensory group an input.
- **Swaps.** N2 and N4 make $10 \cdot |E|$ swap attempts on their edge set; an attempt is rejected if the two edges share a source or a target or if either new edge already exists (for N4, existence is checked against the whole matrix, so interior swaps cannot duplicate a boundary edge). Accepted swaps are not counted by the engine; interior overlap with the connectome (0.111 to 0.121 for N4, 0.089 to 0.095 for N5) is the mixing diagnostic we report.
- **N5.** For each non-sensory source, its interior outputs are moved to randomly chosen free rows among the non-sensory, non-descending groups; its boundary outputs stay in place.
- **Kenyon-cell inputs.** The projection-neuron-to-Kenyon-cell matrix is randomised by degree-preserving swaps in N2, N4 and N5, and by shuffling which Kenyon cells receive each group’s input in N1. The Kenyon-cell-to-group matrix and initial Kenyon-cell-to-output weights are shared.
- **Boundary check.** For N4 and N5 the engine compares every boundary edge with the connectome before calibration and aborts if any differs (maximum absolute difference 0 in every construction).

B.3 Calibration

All experiments here use **global** calibration: W row-normalised to unit absolute input, then a global gain and a Kenyon-cell gain found by bisection so that mean group activity is 0.15 and mean Kenyon-cell activity 0.05 over a fixed stimulus set. Convergence is exact in every condition. Because row normalisation divides each row by its own total input, a boundary edge that is bit-identical before calibration can carry a slightly different share of weight afterwards; all path statistics in Section 4.1 are computed on the calibrated matrices that the runs actually used.

Global calibration matches the mean activity but not its spread: across the first grid the connectome had between-group activity SD 0.163, 46 silent groups of 512 and 2.0 % saturated, against SD 0.080 to 0.086, 8 to 19 silent and under 0.3 % saturated for N1 to N3; N4 and N5 (corrected grid, 40 runs each) have median SD 0.087 (runs 0.081 to 0.095) and median 10.5 and 13 silent groups (runs 4 to 17). Paper-v1 repeated its main endpoints under a distribution-matched calibration that equalises mean, spread and silent fraction, with the same conclusions (Appendix I).

- **global**: W row-normalised to unit absolute input; one global gain and one Kenyon-cell gain found by bisection so mean group activity = 0.15 and mean Kenyon-cell activity = 0.05 over a fixed stimulus ensemble (16 odors $\times$ 2 concentrations, single-channel stimuli, one mixed stimulus). Exact convergence for all conditions (relative error 0).
- **pergroup**: per-group multiplicative gain feedback, 150 iterations (the exploratory-phase scheme). Residuals differ by condition (connectome 0.23, nulls 0.02–0.07).
- **rawglobal**: global gain without structural normalisation; circuit analysis only.
- **distmatch**: structural normalisation plus per-group quantile matching to a fixed log-normal reference profile (mean 0.15, SD/mean 0.8), equalising mean, SD and silent fraction. Achieved: mean 0.148–0.149, SD 0.118, sorted-profile error 0.0013 in all four conditions.

C Evaluation and statistics

Every 50 generations the 16 agents with the highest training fitness per island are evaluated for 8 further lifetimes each in the run’s own ecology. The random-number generator is reset to seed 777 before the batch, so conditions and ablations face the same sequence of generated worlds and the comparison is paired at the level of the batch, not of the individual arena. Evaluation worlds are drawn from the same distribution as training worlds and are not held out. The run-level number is the mean over the 128 lifetimes. Five versions run in parallel: intact, plasticity blocked, olfaction blocked, vision blocked, and all read-outs zeroed. Ablation cost is intact minus ablated. The registered primary endpoint is the time average of this fitness over generations 0 to 550, computed by the trapezoid rule and divided by the generation span; the fitness at the last of these probes is a registered secondary endpoint. Because the probes are 50 generations apart, that last probe is at generation 550, while evolution itself continues to generation 600; wherever we report an end-of-run difference it is measured at generation 550. Drafts up to v3.2 labelled these numbers “generation 600”, which was wrong; the values were always the last probe and are unchanged.

Seeds are the unit of replication. Within an ecology we test paired differences (connectome minus control) with two-sided Wilcoxon signed-rank tests, Holm-corrected across controls, and report medians with percentile bootstrap intervals over resampled seeds (10,000 resamples). Runs of the same seed share their random controls across ecologies, so statements across ecologies average each seed’s ecologies first and test the 10 seed means; the estimand is then the median across seeds of a seed’s equally weighted mean difference. Counts over all 40 runs are descriptive.

For claims that a difference is absent we ask for equivalence rather than a non-significant p-value: the 90 % bootstrap interval of the median paired difference must lie inside $\pm\delta$ with $\delta = 0.10$ fitness units. The bound was chosen after the first grid, when the size of the shortcut effect was known (the smallest shortcut effect we wanted to exclude was 0.20) and before any equivalence interval was computed; it is a claim about what is small relative to that effect, not about what is biologically negligible. Equivalence is claimed only where we state it, and only for the estimand above; a non-significant test elsewhere is reported as no detected difference. Analyses added during AI-assisted review of earlier drafts (Appendix J) are labelled post hoc.

D Pre-registration record and the noise correction

The first grid was run with the heading-noise term written as 0.05ξ with $\xi \sim U(0, 1)$, which adds a steady 0.025 rad per step; because the turn read-out subtracts a running mean of the left–right difference, no network could offset the drift. The error came to light through a question about that term during AI-assisted review of the first draft. Re-evaluating the stored first-grid populations with zero-mean noise showed that connectome populations lost 0.174 fitness when the drift was removed while controls lost 0.00 to 0.04 (10 of 10 seeds, $p_{\text{holm}} = 0.0059$), and that the connectome’s generation-0 lead over N1 and N2 became non-significant; the late overtaking by N1 and N2 was unchanged.

Every experiment after the first grid uses the zero-mean form given in Appendix A. The first grid is reported in Appendix E because it motivated the rest and because the overtaking it shows is the effect that the later experiments explain; its generation-0 result is not used.

The protocols were frozen in this order: PROTOCOL_ECO.md before the first grid, PROTOCOL_MUT.md after its analysis, PROTOCOL_FIX.md (with the boundary-preserving controls added in amendment 1) before the corrected grid, and PROTOCOL_SWAP.md before the intervention runs and before the corrected grid was analysed. The research log in the repository records the exploratory phase that preceded the first protocol, including the design changes made then; none of those affect the frozen experiments reported here.

D.1 Design changes and corrections, in order

The main text reports one correction made after pre-registration (Appendix D). For completeness, this section lists every design change that reached the frozen protocols or the reported results, in the order they were made; the full dated log is RESEARCH_LOG.md in the repository.

when	what	effect on reported results
exploratory phase, before PROTOCOL.md	27 changes to brain construction, calibration and world code (for example: mandatory inclusion of the gustatory-to-dopaminergic path groups; replacement of clustered Kenyon cells by individually wired ones; removal of an energy loophole that let agents survive without eating; a retracted learning-specificity result traced to a numerically unstable calibration)	none of the frozen experiments were run before these changes
after PROTOCOL_ECO.md was frozen, before any run	the setting meant to remove the predator instead enabled an adaptive predator; fixed and re-frozen before the first grid	none
after the first grid	stored genomes were population samples rather than elites, by an indexing error	affects only the post hoc assays of Appendix H, as stated there
after the first grid (AI-assisted review of draft 1)	one-sided heading noise, Appendix D	the first grid's generation-0 result is not used; all later experiments corrected
after the first grid	the standard nulls carry olfactory-to-motor shortcuts, Section 4.1	motivated the corrected grid and the intervention

E First grid

The first grid (pre-registered; 4 ecologies $\times$ 10 seeds $\times$ 600 generations; conditions connectome, N1, N2, N3; noise error present, Appendix D) reproduced the pattern that motivates this paper. Connectome populations started ahead of all three controls in every ecology (12 of 12 comparisons, $p_{\text{holm}} \leq 0.027$) and were then overtaken. At the last probe the column shuffle N1 was ahead in every ecology (p_{holm} 0.006 to 0.041) and, on seed means across ecologies, in all 10 seeds ($p_{\text{holm}} = 0.0059$); the degree-preserving control N2 was ahead in 8 of 10 seeds ($p_{\text{holm}} = 0.0195$). The registered primary endpoint, time-averaged fitness, showed no connectome advantage over either (seed-mean medians -0.10 against N1 and -0.03 against N2). Crossings were sustained in 36 of 40 runs for N1 and 31 of 40 for N2 (Figure 4). Connectome populations also depended on olfaction far less than the controls in all 12 ecology $\times$ control comparisons ($p_{\text{holm}} = 0.0059$ each).

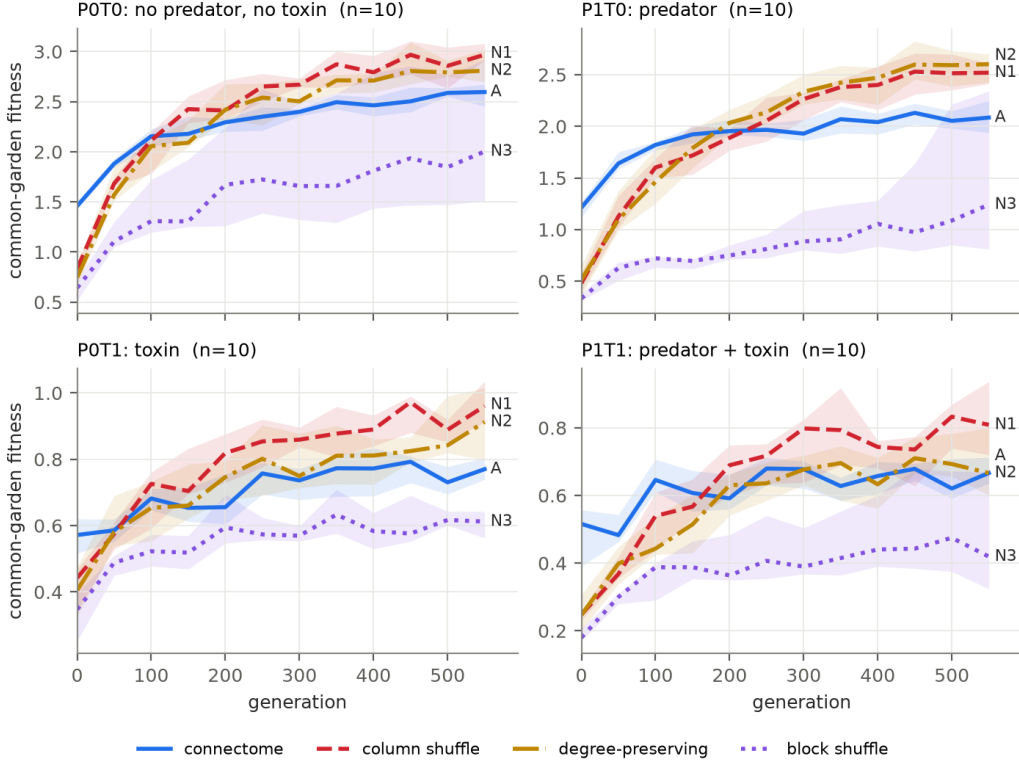


Figure 4: First grid: common-garden fitness over 600 generations in the four ecologies, median and interquartile range over 10 seeds. This experiment was run before the noise correction (Appendix D) and uses controls that carry sensory-to-motor shortcuts.

One explanation can be set aside. In the mutational-neighbourhood assay the fraction of one-step mutants beating their parent's clone distribution was the same for connectome and control genomes (9.6 % against 9.4 % for N1 and 9.8 % for N2 at generation 0; 7.0 % against 7.0 % and 6.4 % at generation 500), with run-level differences whose 95 % intervals all lie inside ± 2.7 percentage points; at the ± 2 percentage-point margin set in that analysis, 12 of the 18 comparisons fall inside and 6 do not, so this is similarity at a coarse resolution rather than demonstrated equivalence. This is a statement about one-step accessibility at this resolution, not about evolvability in general: it does not exclude differences in the size of the best mutations, in multi-step accessibility, or in epistasis. Connectome genomes did lose more to large mutations (10th percentile -0.87 of parent fitness against -0.77 to -0.81 , p_{holm} 0.007 to 0.012 at generation 0).

F Corrected grid: further results

We re-ran the grid with the noise term corrected and with N4 and N5 in place of N3 (PROTOCOL_FIX.md; 40 runs, 5 conditions each, 600 generations). Table 3 gives differences at the last probe.

Against the shortcut-carrying controls the connectome is behind ($p_{\text{holm}} = 0.029$ for N1 and 0.016 for N2 on seed means). Against the boundary-preserving controls the seed-mean difference lies inside the equivalence bound for both, $+0.002$ for N4 and -0.074 for N5, the latter close to the edge. Per ecology the estimates are not uniform and we do not claim equivalence there: in the safe-food ecology N5 is ahead of the connectome by 0.26 ($p_{\text{holm}} = 0.029$) and in the predator-plus-toxin ecology the connectome is ahead of N4 by 0.12, with an interval spanning zero. Ten further seeds in the two T0 ecologies, run under rules fixed beforehand (Supplementary S13), did not replicate that lead (seeds 10 to 19: $+0.031$, 90 % CI -0.068 to $+0.152$; all 20 seeds: -0.104 , -0.208 to $+0.016$, $p_{\text{holm}} = 0.27$). At $n = 20$ the per-ecology intervals for N4 and N5 lie inside ± 0.10 in P1T0 but not

in P0T0; N1 stays ahead in both ($p_{\text{holm}} \leq 0.004$), N2 stays ahead in P1T0 ($p_{\text{holm}} = 0.046$) but its P0T0 lead shrinks to -0.214 (-0.375 to $+0.042$, $p_{\text{holm}} = 0.25$). The registered primary endpoint gives seed-mean medians of -0.03 (N1), -0.02 (N2), $+0.03$ (N4) and -0.01 (N5), none significant, and we report these as no detected difference rather than as equivalence.

Because δ was fixed after the first grid (Appendix C), we also report, post hoc, the smallest symmetric bound each seed-mean interval satisfies (Appendix F): 0.045 for N4 and 0.096 for N5. The N4 verdict therefore holds for any bound down to 0.05; the N5 verdict holds at 0.10 but not at 0.075 and should be read as marginal.

The sensory-strategy difference behaves the same way. The cost of blocking olfaction is 0.03 for the connectome against 0.68 for N1 and 0.64 for N2 ($p_{\text{holm}} = 0.008$), and 0.10 for N4 and 0.17 for N5. The differences from the connectome are -0.07 $[-0.11, -0.02]$ for N4 and -0.11 $[-0.22, +0.02]$ for N5, neither significant after Holm correction across the four controls ($p_{\text{holm}} = 0.13$ for both), and five- to ninefold smaller than the 0.6 differences against N1 and N2. For vision the corresponding differences are $+0.11$ $[+0.03, +0.22]$ against N1 and $+0.20$ $[+0.11, +0.32]$ against N2, both significant, and $+0.03$ $[-0.25, +0.15]$ against N4 and -0.04 $[-0.18, -0.00]$ against N5. With the noise term corrected and the boundary matched, the connectome no longer starts ahead either: the generation-0 seed-mean difference against N4 is -0.18 ($p_{\text{holm}} = 0.15$), with N4 slightly ahead.

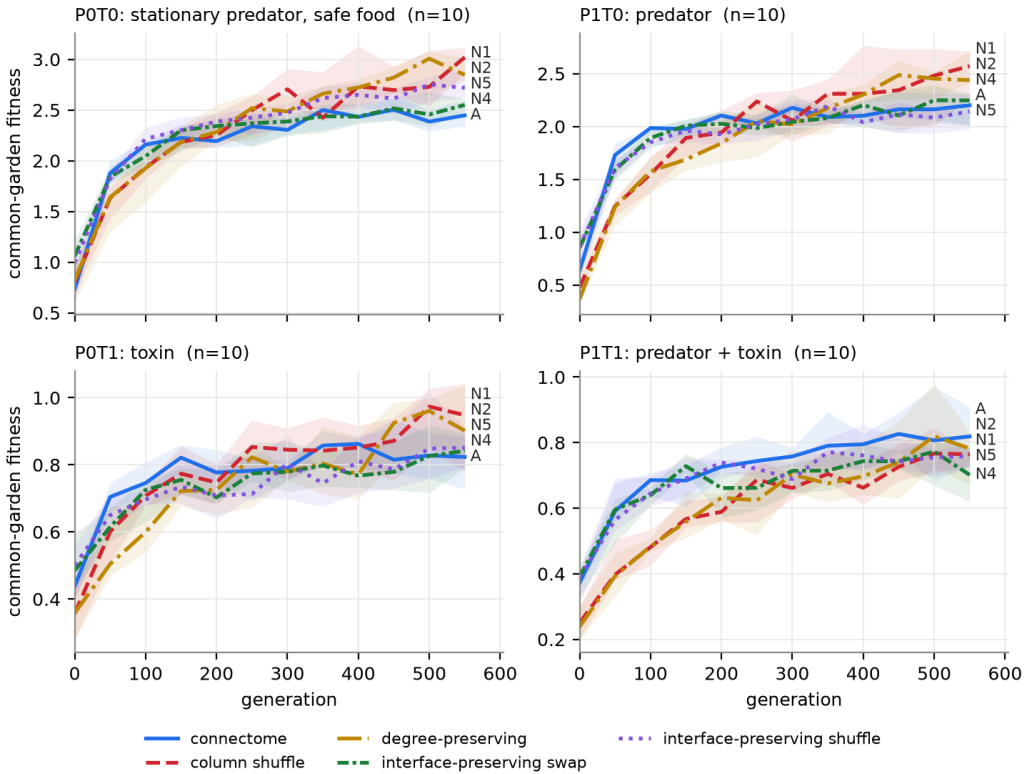


Figure 5: Corrected grid: connectome, shortcut-carrying controls (N1, N2) and boundary-preserving controls (N4, N5).

F.1 Sensitivity of the equivalence verdicts to the bound

The bound $\delta = 0.10$ was fixed after the first grid (Appendix C). Here each registered 90 % interval of the corrected grid (connectome minus control, last probe) is read against other bounds. The smallest symmetric bound an interval satisfies is the larger of its two ends in absolute value. The intervals are read from the file Table 3 is taken from (`results/equivalence.txt`) by `analyze_delta_sens.py`; nothing is recomputed. Equivalence is claimed in the paper only on seed means; the per-ecology rows are shown to make the heterogeneity visible, not as claims.

cell	control	median	90 % CI	smallest bound	$\delta =$ 0.05	0.075	0.10	0.15	0.20
seed mean	N4	+0.002	[-0.026, +0.045]	0.045	yes	yes	yes	yes	yes
seed mean	N5	-0.074	[-0.096, +0.002]	0.096	no	no	yes	yes	yes
P0T0	N4	-0.064	[-0.255, +0.002]	0.255	no	no	no	no	no
P0T0	N5	-0.264	[-0.368, -0.179]	0.368	no	no	no	no	no
P1T0	N4	+0.021	[-0.026, +0.038]	0.038	yes	yes	yes	yes	yes
P1T0	N5	+0.067	[-0.067, +0.090]	0.090	no	no	yes	yes	yes
P0T1	N4	-0.031	[-0.116, +0.085]	0.116	no	no	no	yes	yes
P0T1	N5	-0.015	[-0.102, +0.061]	0.102	no	no	no	yes	yes
P1T1	N4	+0.120	[-0.064, +0.283]	0.283	no	no	no	no	no
P1T1	N5	+0.070	[-0.009, +0.205]	0.205	no	no	no	no	no

On seed means the N4 verdict holds for every bound down to 0.05. The N5 verdict holds at the registered 0.10 but not at 0.075: it is marginal, and the paper says so. Per ecology the smallest bound ranges from 0.038 (N4, P1T0) to 0.368 (N5, P0T0).

F.2 Distribution-matched calibration (D-06)

Frozen before the runs: protocol PROTOCOL_DIST2.md (0124b50ad5345279), analysis analyze_dist2.py (d12b803c141faa1e, unchanged at analysis), the corrected grid's frozen engine evo_fix.py and flags with --homeo distmatch in place of global. All 40 runs (4 ecologies $\times$ seeds 0–9) finished; each configuration equals its global-calibration counterpart apart from run name and calibration. Output results/dist2_key.json.

Manipulation check (registered: at least 36 of 40 runs with $|SD - SD_A| \leq 0.01$, $|silent - silent_A| \leq 10$ and $|saturated - saturated_A| \leq 0.02$ for N4 and N5): 40/40 runs $\rightarrow$ PASSED. Calibrated activity by condition, medians over the 40 runs:

condition	between-group SD	silent groups	Kenyon-cell mean
A	0.118	7	0.050
N1	0.118	3	0.050
N2	0.118	6.5	0.050
N4	0.118	4.5	0.050
N5	0.118	4.5	0.050

Under global calibration the same quantities were SD 0.163 / 46 silent groups for the connectome against 0.080–0.087 / 8–19 for the controls (main text); distribution matching removes that difference. N1's Kenyon-cell mean ranged 0.038–0.050 (a known limitation of the scheme for the column shuffle).

P1 and P2 — seed means over four ecologies, median with 90 % bootstrap interval:

contrast	global (registered grid)	distribution-matched	verdict
A – N4	+0.002 [–0.026, +0.045]	+0.055 [+0.035, +0.097]	EQUIVALENT
A – N5	–0.074 [–0.096, +0.002]	–0.009 [–0.076, +0.046]	EQUIVALENT
A – N1	–0.224 [–0.337, –0.103]	–0.274 [–0.306, –0.123]	BEHIND
A – N2	–0.199 [–0.321, –0.133]	–0.146 [–0.231, –0.037]	BEHIND

S1 — change caused by calibration, seed mean of (A – N) under distribution matching minus (A – N) under global:

control	change	verdict
N1	+0.010 [–0.112, +0.072]	UNDETERMINED
N2	+0.101 [+0.016, +0.133]	CHANGED
N4	+0.062 [–0.002, +0.117]	UNDETERMINED
N5	+0.076 [–0.017, +0.128]	UNDETERMINED

S2 — N5 ahead of the connectome in P0T0 under distribution matching: A – N5 = –0.066 [–0.320, +0.124] → NOT REPLICATED.

Per ecology (descriptive), A minus control, median [90 % CI]:

ecology	control	global	distribution-matched
P0T0	N1	–0.621 [–0.661, –0.469]	–0.437 [–0.563, –0.235]
P0T0	N2	–0.403 [–0.421, –0.308]	–0.478 [–0.677, –0.052]
P0T0	N4	–0.064 [–0.255, +0.002]	+0.139 [–0.128, +0.230]
P0T0	N5	–0.264 [–0.368, –0.179]	–0.066 [–0.320, +0.124]
P1T0	N1	–0.312 [–0.441, –0.197]	–0.489 [–0.664, –0.275]
P1T0	N2	–0.338 [–0.530, –0.074]	–0.164 [–0.483, –0.013]
P1T0	N4	+0.021 [–0.026, +0.038]	+0.051 [–0.120, +0.144]
P1T0	N5	+0.067 [–0.067, +0.090]	+0.041 [–0.204, +0.214]
P0T1	N1	–0.159 [–0.205, +0.010]	–0.096 [–0.182, +0.010]
P0T1	N2	–0.017 [–0.333, +0.005]	–0.020 [–0.122, +0.096]
P0T1	N4	–0.031 [–0.116, +0.085]	+0.073 [–0.066, +0.104]
P0T1	N5	–0.015 [–0.102, +0.061]	+0.036 [–0.086, +0.089]
P1T1	N1	+0.054 [–0.054, +0.127]	+0.063 [–0.040, +0.099]
P1T1	N2	+0.015 [–0.132, +0.137]	+0.104 [–0.091, +0.111]
P1T1	N4	+0.120 [–0.064, +0.283]	–0.012 [–0.099, +0.073]
P1T1	N5	+0.070 [–0.009, +0.210]	–0.027 [–0.066, +0.078]

F.3 Extension to 20 seeds in two ecologies

Seeds 10 to 19 of the two T0 ecologies were run with the corrected grid’s frozen engine and flags after rules R1 to R3 were written (research log 46; `analyze_n20.py`). Every one of the 40 runs has the same configuration as seed 0 apart from name and seed. Endpoint: fitness at the last common-garden probe (generation 550); connectome minus control, median with 90 % bootstrap interval; `p_holm` from Wilcoxon signed-rank tests at $n = 20$, Holm-corrected across the four controls within each ecology.

ecology	control	seeds 0–9	seeds 10–19	n = 20	p_holm (n = 20)	inside ± 0.10 (R1)
P0T0	N1	–0.621 [–0.661, –0.469]	–0.304 [–0.513, –0.161]	–0.491 [–0.624, –0.298]	0.001	no
P0T0	N2	–0.403 [–0.421, –0.308]	+0.100 [–0.068, +0.332]	–0.214 [–0.375, +0.042]	0.248	no
P0T0	N4	–0.064 [–0.255, +0.002]	+0.090 [–0.109, +0.217]	–0.011 [–0.114, +0.080]	0.841	no
P0T0	N5	–0.264 [–0.368, –0.179]	+0.031 [–0.068, +0.152]	–0.104 [–0.208, +0.016]	0.265	no
P1T0	N1	–0.312 [–0.441, –0.197]	–0.417 [–0.522, +0.067]	–0.333 [–0.482, –0.197]	0.004	no
P1T0	N2	–0.338 [–0.530, –0.074]	–0.101 [–0.335, +0.103]	–0.203 [–0.430, –0.032]	0.046	no
P1T0	N4	+0.021 [–0.026, +0.038]	–0.027 [–0.168, +0.050]	+0.003 [–0.042, +0.029]	1.000	yes
P1T0	N5	+0.067 [–0.067, +0.090]	+0.003 [–0.175, +0.144]	+0.056 [–0.097, +0.090]	1.000	yes

R2 (replication of N5’s lead in P0T0 in seeds 10 to 19): A – N5 = +0.031 [–0.068, +0.152] → **NOT REPLICATED.**

R3 (two-ecology pool, seed means over P0T0 and P1T0, n = 20; a different estimand from Table 3’s four-ecology seed means):

control	median [90 % CI]	seeds with connectome ahead	inside ± 0.10
N1	–0.401 [–0.497, –0.241]	3/20	no
N2	–0.155 [–0.273, –0.060]	4/20	no
N4	0.000 [–0.087, +0.083]	10/20	yes
N5	–0.059 [–0.093, +0.018]	7/20	yes

Post hoc checks on the extension (not registered; analyze_posthoc_n20.py). The nulls of seeds 10 to 19 were built like those of seeds 0 to 9: the olfactory-to-descending direct share of the calibrated weights the runs used was 10.41 % (median) for N2 against 10.63 % in seeds 0 to 9, and 0.01 % for N4 and N5. Across the 20 null realisations per ecology this share ranged from about 5 to 17 % but did not predict the null’s lead over the connectome (Spearman $|\rho| \leq 0.21$, $p \geq 0.38$ for N1 and N2 in each ecology; pooled $\rho = -0.00$ and -0.09). The interventions changed the connectome’s share from 0.01 % to 1 to 10 %, a range these realisations do not cover. A rough plug-in simulation from the observed differences suggests that per-ecology equivalence in P0T0 would need on the order of 100 seeds for N4 and is unlikely for N5 at any sample size, so we did not extend further.

F.4 Additional analyses (post hoc)

Full outputs: `results/eco_analysis.txt` (registered tests), `results/eco_cross.txt` (crossing, cross-ecology, H3), `results/eco_pooled.txt` (across-ecology summaries), `results/mut_analysis.txt` (mutational neighbourhood with confidence intervals), `results/revision_analysis.txt` (sustained crossing, fitness components, absolute ablation costs, population samples against top 16), `results/eco_calib.txt` (activity diagnostics per run).

G Intervention and dose-response: further details

G.1 Intervention

The pre-registered intervention (`PROTOCOL_SWAP.md`; 20 runs in the two ecologies with the clearest overtaking, corrected noise) rewires the shortcut in both directions while preserving degrees and Dale signs.

Adding shortcuts to the connectome raised evolved fitness in every seed: AS minus connectome at the last probe was +0.44 on seed means (10 of 10 seeds, $p = 0.002$; +0.445 in P0T0 and +0.475 in P1T0), and time-averaged fitness +0.26 (10 of 10, $p = 0.002$). The effect grows over evolution: at generation 0 the difference is +0.04 ($p = 0.32$, 90 % CI -0.04 to +0.18, wider than the equivalence bound, so an initial effect is not excluded), and the increase from generation 0 to the last probe is +0.40 (10 of 10 seeds, $p = 0.002$). The intervention also transferred the sensory strategy: the cost of blocking olfaction rose from 0.15 to 0.99 ($p = 0.002$), close to the shortcut-carrying control's 0.84.

Removing the shortcut from the degree-preserving control cost it 0.43 fitness in the safe-food ecology ($p = 0.004$) and an estimated 0.00 in the predator ecology (the difference between ecologies is -0.36, $p = 0.027$). In neither ecology could we detect a difference between the shortcut-free control and the connectome (P0T0 -0.11, 90 % CI -0.23 to +0.25; P1T0 +0.05, 90 % CI -0.24 to +0.23), and its olfactory dependence fell from 0.84 to 0.58 ($p = 0.027$). In this experiment the degree-preserving control's own advantage over the connectome was +0.55 in P0T0 ($p = 0.006$) and +0.10 in P1T0 ($p = 0.23$), smaller in the predator ecology than the corrected grid's +0.34: controls are drawn from a single random stream in the order the conditions are listed, so the intervention runs use different N2 realisations than the corrected grid, and this is the size of the variation between realisations.

Shortcut-creating rewiring is therefore sufficient to produce the control's advantage and to transfer its olfactory strategy, and shortcut-removing rewiring removes the advantage where it was clear. Because both interventions also rewire edges that are not shortcuts, path shortening is the proposed mechanism rather than an isolated variable. Section 4.4 reports the rewiring control this calls for, and it does not come out clean.

G.2 Dose-response and sham

The intervention in Section 4.3 moved the shortcut share between its two extremes and changed every rewired edge at once, so it left two questions open: whether the effect follows the amount of shortcut, and whether it needs the shortcut at all. We pre-registered both (`PROTOCOL_DOSE.md`; 2 ecologies $\times$ 10 seeds $\times$ 600 generations) and ran four graded doses, raising the connectome's olfactory-to-descending share to at least 1, 3, 5 and 10 % by degree-preserving swaps, against a sham that performs 101 swaps on edge pairs whose source is not an olfactory receptor and whose target is not a descending group, leaving the shortcut share unchanged at 0.0001. Each seed draws its own swaps, so the share reached and the swaps it took vary across the ten seeds: 1.0 to 1.7 % (3 to 27 swaps), 3.0 to 4.6 % (11 to 48), 5.0 to 5.8 % (25 to 58) and 10.0 to 10.6 % (68 to 129), with no overlap in share between doses. The sham's 101 is the full transplant's swap count on a fixed stream and is the same in every seed.

The comparison that separates the two variables is the full dose against the sham: they rewire comparable amounts (the full dose took 68 to 129 swaps depending on the seed, mean 98.5, against the sham's 101) and differ by four orders of magnitude in shortcut share. The full dose is ahead of the sham by 0.534 on seed means (90 % CI +0.373 to +0.693, 10 of 10 seeds; +0.525 in the safe-food ecology with 10 of 10 seeds and +0.543 in the predator ecology with 9 of 10). The swap counts are not matched seed by seed, and that strengthens the comparison rather than weakening it: in the seven seeds where the full dose used fewer swaps than the sham it was ahead in all seven, by 0.612 on average, against 0.352 in the three where it used more, and across seeds the gap does not grow with

the difference in swap count ($r = -0.16$). The number of degree-preserving swaps therefore does not account for the effect; what those swaps land on does.

Fitness also rose across the graded doses. On seed means over both ecologies it went 2.37, 2.54, 2.69, 2.75, 2.93 from the connectome to the 10 % dose (Page's L , $z = 4.68$, $p = 1.4 \times 10^{-6}$; per-seed Spearman correlation positive in 9 of 10 seeds, mean $+0.74$), and the trend held separately in both ecologies ($p = 1.5 \times 10^{-5}$ and 0.004). Dependence on olfaction rose with it: the fitness lost when olfaction is blocked went from 0.28 to 1.26 in the safe-food ecology and from 0.08 to 0.95 in the predator ecology ($p = 3.4 \times 10^{-5}$ and 5.6×10^{-7}). At generation 0 the doses are indistinguishable ($p = 0.98$), so the graded effect is a product of evolution rather than of the initial wiring. We ran a sham only at the full swap count, so shortcut share and swap count rise together across the intermediate doses and the shape of the curve between them cannot separate the two; only its endpoint has a sham. Shams paired with each intermediate dose within the same run would settle the shape.

The sham itself did not behave as we predicted. We registered that sham minus connectome would fall inside the ± 0.10 equivalence bound, which would have shown that rewiring by itself does nothing. It does not: on seed means the difference is $+0.034$ with a 90 % interval of -0.062 to $+0.123$, which crosses the bound. The two ecologies also disagree in sign, the sham being ahead by 0.144 in the safe-food ecology (90 % CI $+0.072$ to $+0.222$) and behind by 0.077 in the predator ecology (90 % CI -0.257 to $+0.087$), so this is an unstable effect rather than a small consistent one. Following the interpretation registered for this outcome, we report that some part of the advantage follows from degree-preserving rewiring in general. It is a small part next to the 0.534 that the shortcut adds at the same swap count, and we cannot say from these runs what it consists of.

The registered equivalence prediction was not supported. An unregistered analysis of final-generation in-run fitness does meet the equivalence criterion (Appendix G), and does not change that conclusion.

Taken with Section 4.3: at a fixed amount of degree-preserving rewiring, whether that rewiring creates a sensory-to-motor shortcut decides most of the difference, and it brings the olfactory strategy with it. Rewiring that creates no shortcut is not inert, but what it does is smaller and does not hold its sign across ecologies.

G.3 The sham equivalence test on an unregistered outcome

PROTOCOL_DOSE.md registered the common-garden probe of Appendix C as the outcome, as PROTOCOL_ECO.md does for every experiment in this paper. On that outcome the registered equivalence prediction for the sham fails: sham minus connectome is $+0.034$ on seed means, 90 % CI -0.062 to $+0.123$, crossing the ± 0.10 bound (Section 4.4).

The engines also log an in-run fitness at each generation, evaluated in the training worlds rather than in the common-garden batch, and the last of those is generation 599. That outcome was not registered. On it the same difference is -0.022 , 90 % CI -0.090 to $+0.047$, which lies inside the bound and would have supported the prediction. We report this because it is recoverable from the released logs and because the direction of the disagreement is the one that flatters us; it does not govern the registered test, which is the probe.

The disagreement is confined to the sham. Under the unregistered outcome the graded series still rises monotonically (Page's L , $z = 5.12$, $p = 1.5 \times 10^{-7}$, per-seed Spearman positive in 10 of 10 seeds) and the generation-0 null still holds ($p = 0.98$). The swap-matched comparison of Section 4.4 is likewise unaffected in sign or approximate size.

We did not investigate why the two outcomes differ for the sham specifically. The probe resets its random stream and evaluates 16 agents per island for 8 lifetimes in a fixed sequence of worlds, while the in-run number is the population's fitness in the worlds it was selected in; a condition whose advantage is partly specific to the worlds it evolved in would show the two diverging, but we have not tested that here.

G.4 Boundary-targeted sham: full results and execution record

Frozen before the runs (research log 54): protocol PROTOCOL_BSHAM.md, engine evo_bsham.py (007fa61ff2c6f5c5), analysis analyze_bsham.py (cbb7f9e1c7aa6192, unchanged at analysis). Output results/bsham_key.json. All 20 runs finished; the analysis's pre-checks passed (config-

urations equal apart from run, seed and predator speed; 80 boundary-sham constructions complete, shortcut share unchanged, degrees kept; $|w(Z \rightarrow DN)|/|w(ORN \rightarrow X)|$ medians 0.92 to 1.05).

quantity (seed mean over P0T0 and P1T0, n = 10)	mean	90 % CI	registered rule
P: AS10 – BS10, last probe	+0.600	[+0.414, +0.797]	lower bound > +0.10 → shortcut-specific
R: AS10 – BS10, fresh worlds (4 × 8 lives, generation-599 elites)	+0.639	[+0.490, +0.792]	same category as P → robust
S2: AS10 – BS10, P0T0 / P1T0	+0.568 / +0.631	[+0.315, +0.832] / [+0.428, +0.831]	descriptive
S3: BS10 – A	+0.007	[-0.070, +0.086]	±0.10 reference: inside
S4: olfaction-ablation cost AS10 / BS10	1.175 / 0.354	difference +0.821 [+0.563, +1.079]	descriptive
S5: AS10 – A	+0.606	[+0.435, +0.798]	lower bound > 0 → replicated

S1 (ASk – BSk increases with dose): 0.158, 0.277, 0.357, 0.600 for 1, 3, 5, 10 %; Page’s L = 279, z = 3.18, p = 7.4×10^{-4} → supported.

Execution record. Home workstation (M2 Max, 32 GB), one run at a time. Three runs failed with GPU out-of-memory (bsham_P0.2_T0.0_s1 twice, bsham_P0.0_T0.0_s2, bsham_P0.0_T0.0_s5); the causes were the reproduction step holding several population-sized weight copies (peak 30.7 to 33 GB) and, once, another service loading a 12 GB model on the same GPU. One completed attempt was deleted by a driver that misread the log’s last line, and one attempt was stopped deliberately to test the memory fix. Every affected run was re-run from scratch with identical flags; no outcome was inspected before re-running (research logs 55 to 57 and 62). The first four runs (s0 both ecologies, s1 P0T0, s2 P1T0) used `evo_bsham.py`; the other 16 used `evo_bsham_mem.py` (f142f287e671b147), which adds six forced evaluations in reproduction and is otherwise identical; with probes off it was bit-identical to `evo_bsham.py` on the CPU (peak memory 23.8 GB instead of 30.7 to 33 GB). As for every engine in this project, re-running a seed reproduces the constructions and generation 0 exactly and later generations only in distribution (Section on data and code).

H Further checks

H.1 Assays

Mutational neighbourhood. For each run, checkpoint (generations 0, 250, 500) and stored genome, 32 unmutated clones and 32 one-step mutants at 0.5×, 1× and 2× the mutation magnitudes, each evaluated 4 times in the home ecology. The clones give each parent’s noise distribution; the reported statistic is the fraction of mutants exceeding the 95th percentile of their parent’s clones, which is 5 % when mutations have no effect, together with the tails of the mutant-minus-clone difference.

Turning-noise control. Generation-0 and generation-500 population samples of all 40 first-grid runs re-evaluated in their home ecology with the original and with zero-mean heading noise.

Interior necessity. Generation-500 population samples of the corrected grid re-evaluated intact, with interior weights zeroed, and with interior weights randomly permuted within the interior mask. Losses are expressed as (intact – perturbed)/intact per run; the smallest intact fitness over all runs and conditions was 0.44, so the ratio is well behaved.

H.2 Interior perturbation

Re-evaluating generation-500 populations of the corrected grid with the interior disturbed shows that they depend on it: zeroing interior weights costs 18 to 23 % of fitness and randomly permuting them

costs 46 to 55 %, in every condition, with no detected difference between the connectome and any control (all $p \geq 0.16$). This rules out the possibility that the evolved agents run on the sensory-to-motor boundary alone, which would have made the comparison in Section 4.2 uninformative. It does not show that the task could not be solved by a simpler reflex; it shows that these agents did not solve it that way.

H.3 What survives from the first experiment

Of the first grid’s findings, the connectome’s generation-0 advantage does not survive: it rested on the biased turning noise together with the boundary difference, and is no longer detected once both are addressed. The overtaking by N1 and N2 survives, and Section 4.3 accounts for it.

Two observations hold in both grids and are reported with their metrics in Appendix I. We detected no odor-specific associative learning in any condition: in an odor-reversal assay crossed with plasticity blocking, the interaction never reached significance, although plasticity itself was worth 0.09 to 0.30 fitness units. With 10 seeds a small interaction cannot be excluded. Randomised networks also stayed structurally distinct from the connectome throughout: after 250 generations their support overlap with it had moved by at most 0.003, the ipsilateral share of weight was 0.77 in the connectome against 0.50 in N1 and N2, and reciprocal edges 0.43 against 0.05 to 0.12. Over the same period weights moved by 0.5 to 3 % of their initial spread while gains, leaks and read-outs moved by about 6 to 11 initial standard deviations (results/evolution_analysis.txt, the three calibration schemes of paper-v1). We report that as the measured pattern of change; attributing the adaptation to those parameters would need a freezing experiment we did not run.

I Earlier model version (paper-v1)

The results in this appendix come from paper-v1, the first pre-registered experiment (PROTOCOL.md: 4 conditions $\times$ 10 seeds $\times$ 300 generations in one ecology with toxic probability 0.5, toxic-bite loss 0.4 and a predator pursuing at 0.1·v_max; common garden every 25 generations), run before the ecology grids and before the noise correction. Stored genomes there are also population samples rather than elites. It is included for completeness; the main text does not rest on it.

Circuit properties of initial brains, 4 conditions $\times$ 10 seeds per scheme:

- Kenyon-cell odor decorrelation favours the connectome under **pergroup** (10/10 seeds, $p_{\text{holm}} = 0.006$) and favours the nulls under **global** and **rawglobal** (0–3/10 seeds).
- Dopaminergic responses to sugar and malaise favour the connectome under **global** (10/10, $p_{\text{holm}} = 0.006$), show no consistent difference under **pergroup**, and favour the nulls under **rawglobal**.
- Mushroom-body learning specificity shows no significant difference in any scheme.
- Looming-escape turn sign is correct in the connectome in 10/10 seeds and beats all three nulls under **global** and **pergroup** ($p_{\text{holm}} = 0.006$). Under **rawglobal** it beats only N3 ($p_{\text{holm}} = 0.006$; vs N1 $p = 0.275$, vs N2 $p = 0.129$). Laterality is thus the most calibration-robust property, but not significant against every null in every scheme.

An earlier internal result of a 43-fold learning-specificity advantage was traced to a numerically sensitive calibration and is **retracted**; under the robust schemes the ratio is 1.6–3.0 with no significant difference between conditions.

I.1 An innate head start and a different evolved sensory strategy

Pre-registered under **global** and **pergroup**; repeated post hoc under **distmatch**.

endpoint	global	pergroup	distmatch (post hoc)
generation 0	A > N1, N2, N3 (10/10, 10/10, 10/10), $p_{\text{holm}} 0.006$	A > all (9/10, 10/10, 10/10), p_{holm} 0.006–0.008	A > all (10/10, 9/10, 10/10), $p_{\text{holm}} 0.006$

endpoint	global	pergroup	distmatch (post hoc)
olfaction ablation	A -0.03 vs $+0.31$ / $+0.34$ / $+0.30$; p_holm 0.006–0.014	A -0.08 vs $+0.16$ / $+0.18$ / $+0.22$; p_holm 0.006–0.008	A -0.10 vs $+0.25$ / $+0.32$ / $+0.28$; p_holm 0.006–0.008
vision ablation	A $+0.30$ vs $+0.13$ / $+0.14$ / $+0.07$; p_holm 0.06–0.11	A $+0.33$ vs $+0.09$ / $+0.13$ / $+0.17$; p_holm 0.041	A $+0.47$ vs $+0.17$ / $+0.13$ / $+0.05$; p_holm 0.012
generation 275	no difference vs N1, N2 (p_holm 1.0); vs N3 p_holm 0.059	no difference vs N1, N2; vs N3 p_holm 0.111	no difference vs N1, N2; vs N3 p_holm 0.252
AUC (primary)	A > N1, N3 (p_holm 0.006); N2 n.s.	A > N3 only (p_holm 0.012)	A > N3 only (p_holm 0.012)

Median generation-0 differences are $+0.16$ to $+0.37$ fitness units. The head start and the sensory-strategy divergence replicate in all three calibrations, including the one that matches the full activity distribution, so they are not explained by marginal activity statistics. The cumulative endpoint is not robust: the degree-preserving null N2 is never significantly worse on AUC or final fitness.

I.2 No evidence that the nulls’ olfactory reliance is associative learning (post hoc)

On generation-250 population samples the reversal $\times$ plasticity interaction is not significant in any condition or calibration (p_holm 0.39–1.0), and extra toxic bites after an odor reversal do not differ. A plasticity benefit of ≈ 0.09 – 0.30 exists in all conditions but is not odor-specific. We therefore find **no evidence** that the nulls’ olfaction dependence is mediated by odor-specific associative learning; with 10 seeds and the observed confidence intervals a smaller interaction cannot be excluded.

Transfer: with a wider odor plume, connectome population samples outperform N1 and N3 under **global** (p_holm = 0.029); under **distmatch** no transfer difference reaches significance, and higher toxicity shows none in any scheme.

I.3 The head start is amplified by, but not reducible to, the engineered reflex (post hoc)

Removing the innate feeding read-out reduces the generation-0 advantage from $+0.15$... $+0.31$ to $+0.04$... $+0.08$, still significant in all schemes (7–10/10 seeds, p 0.006–0.029). Blinding alone leaves it near its original size ($+0.13$... $+0.30$). Removing both vision and the read-out leaves $+0.03$... $+0.07$ (8–10/10 seeds, $p \leq 0.029$). The random-walk control, with all read-outs zeroed, gives a difference of exactly 0.000 (0/10 seeds) in every scheme, confirming the differences arise through the behavioural pathway and not through the evaluation itself.

I.4 Convergence with the degree-preserving null is not a ceiling artifact (post hoc)

Generation-250 population samples re-evaluated in harder worlds without further evolution: at predator speed 0.18 the connectome still beats N1 and N3 (global $+0.228$ vs N1, 9/10, $p = 0.010$; distmatch $+0.095$ vs N1, 8/10, $p = 0.027$ and $+0.342$ vs N3, 8/10, $p = 0.027$) but is never significantly different from N2 at any difficulty or in any scheme. At the most extreme toxicity level the connectome advantage disappears in all schemes.

I.5 Negative result: neuron-level central complex

A module of 1,041 real central-complex neurons (EPG, PEG, PEN, Delta7, PFN, h Δ B, FC2, PFL, plus GLNO/LNO/ER interface neurons with measured connectivity) reproduces heading tracking of a rotating visual cue (phase-tracking ratio 1.003; 93 % of EPG variance on a two-dimensional ring). Across $\approx 1,500$ pathway-gain settings from random search and an evolution strategy we found none that sustains heading in darkness, integrates angular velocity, or supports decodable path integration, for the connectome and the shuffled control alike. We report this as a limitation of our rate-model formulation, not as a statement about the connectome.

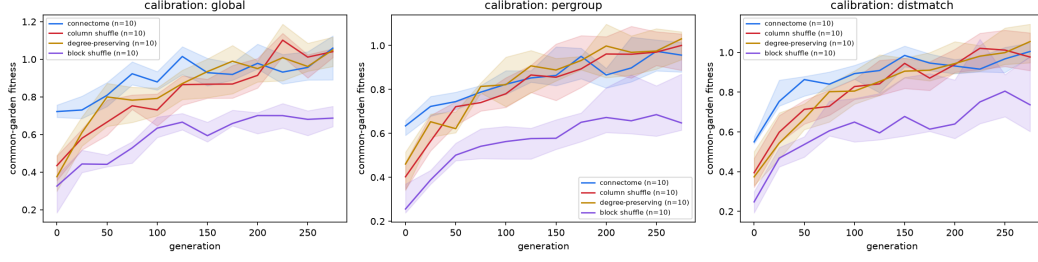


Figure 6: Common-garden fitness over 300 generations in paper-v1, per calibration scheme (median and interquartile range over 10 seeds).

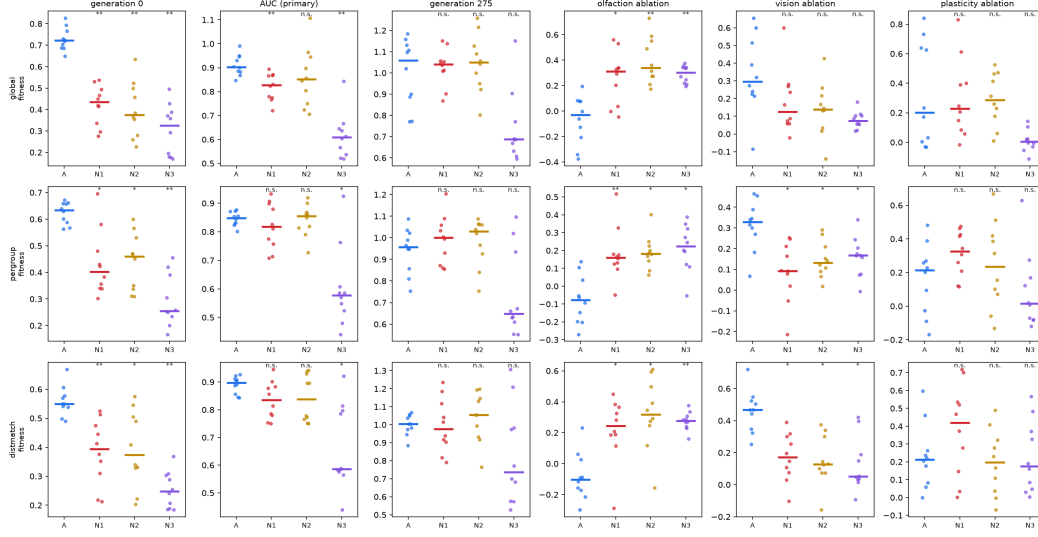


Figure 7: Paper-v1 endpoints per seed: generation 0, AUC (primary), generation 275, and ablation costs.

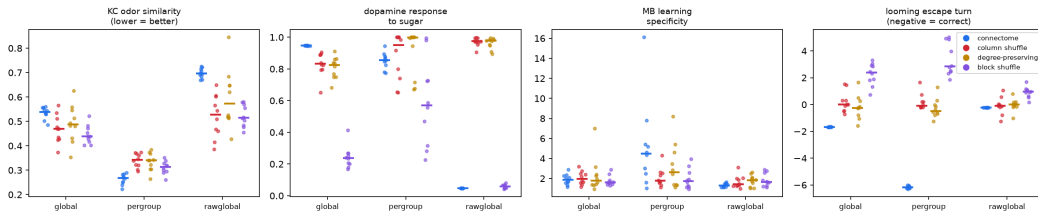


Figure 8: Circuit metrics before evolution under three calibration schemes.

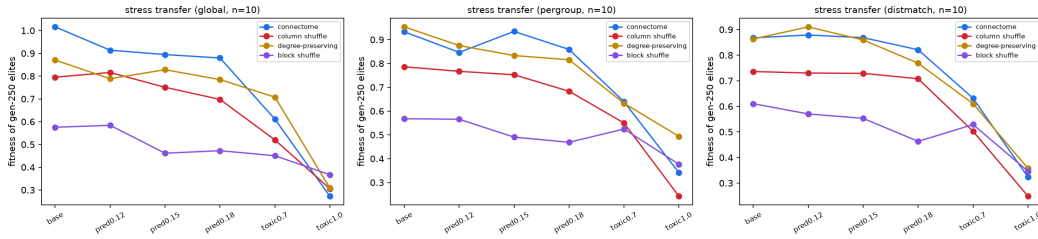


Figure 9: Generation-250 samples of paper-v1 tested in harder worlds without further evolution.

J Use of AI tools

The simulation engines, analysis and figure scripts were written with an AI coding assistant (Claude Code, Anthropic), and the manuscript was drafted with the same assistant from the author's instructions, the stored result files and the dated research log. Two further language models (Gemini, Google; GPT, OpenAI) were used to critique drafts and analyses; their comments are referred to in the text as AI-assisted reviews, not peer review, and the analyses added in response are labelled post hoc. The mutational-neighbourhood assay, the turning-noise control, the equivalence analysis and the interior-necessity assay were added in response to them, and a question raised in one of them led to the discovery of the noise error (Appendix [D](#)). Every number reported here is regenerated from the stored run outputs by scripts in the repository (the data availability statement). The author designed and directed the study, made every decision recorded in the research log, and is responsible for the final text.